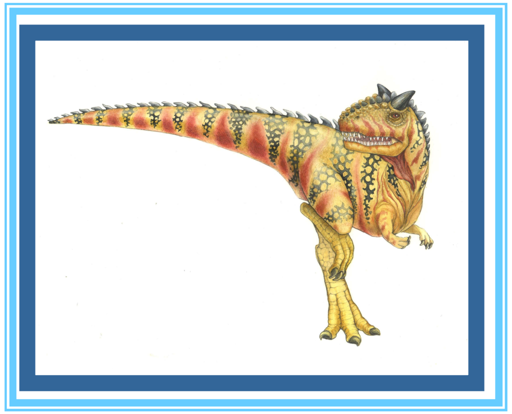
Chapter 1: Introduction



What is an Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware
  - Manages and controls hardware
  - Provides a basic for application programs
- Operating system goals:
  - Execute user programs and make solving user problems easier
  - [User View] Make the computer system convenient to use
    - The operating system is designed mostly for **ease of use.**
  - [System View] Use the computer hardware in an efficient manner
    - Increase resource utilization

Computer System Structure

- Computer system can be divided into four components
  - Hardware – provides basic computing resources
    - CPU, memory, I/O devices
  - Operating system
    - Controls and coordinates use of hardware among various applications and users
  - Application programs – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, web browsers, database systems, video games, compilers, …
  - Users
    - People, machines, other computers

Four Components of a Computer System

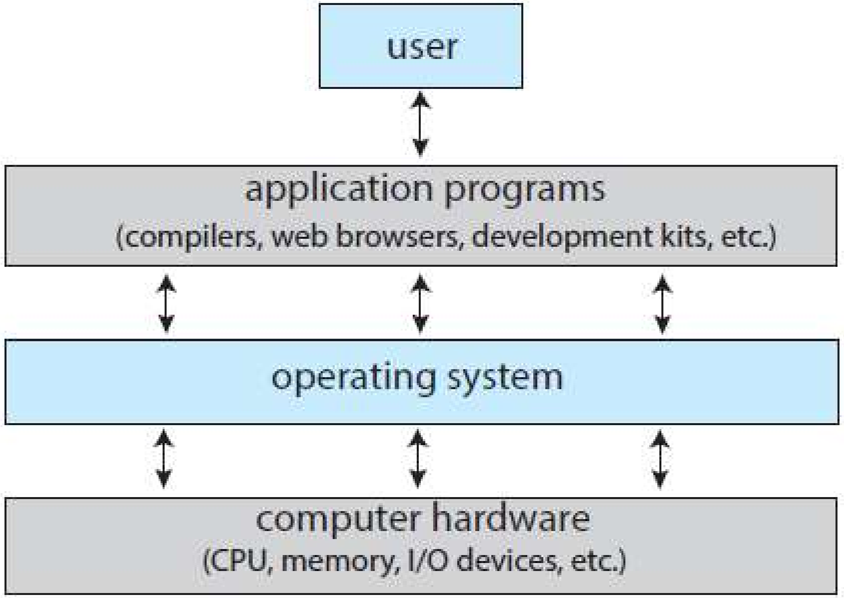


Figure 1.1 Abstract view of the components of a computer system.

Operating System – System View

- OS is a **resource allocator**
  - Manages all resources
  - Decides between conflicting requests for efficient and fair resource use
- OS is a **control program**
  - Controls execution of programs to prevent errors and improper use of the computer

Operating System Definition

- No universally accepted definition
- “Everything a vendor ships when you order an operating system” is good approximation
  - But varies wildly
- “The one program running at all times on the computer” is the **kernel**. Everything else is either a system program (ships with the operating system) or an application program
- System programs (system utilities)
  - Associated with OS but are not part of the kernel
  - Provide a convenient environment for program development and execution
  - Editor, compilers, debuggers, shell, etc.

System software is software designed to provide a platform for other software.

Application software is software that allows users to do user-oriented tasks.

What is OS?

- OS is a program that manages .
- OS is designed to be and
- Four major components of computer system
- We can view OS as
- What is “The one program running at all times on the computer”?

Computer-System Organization

- A modern general-purpose computer system consists of one or more CPUs and a number of device controllers connected through a common bus that provides access between components and shared memory.

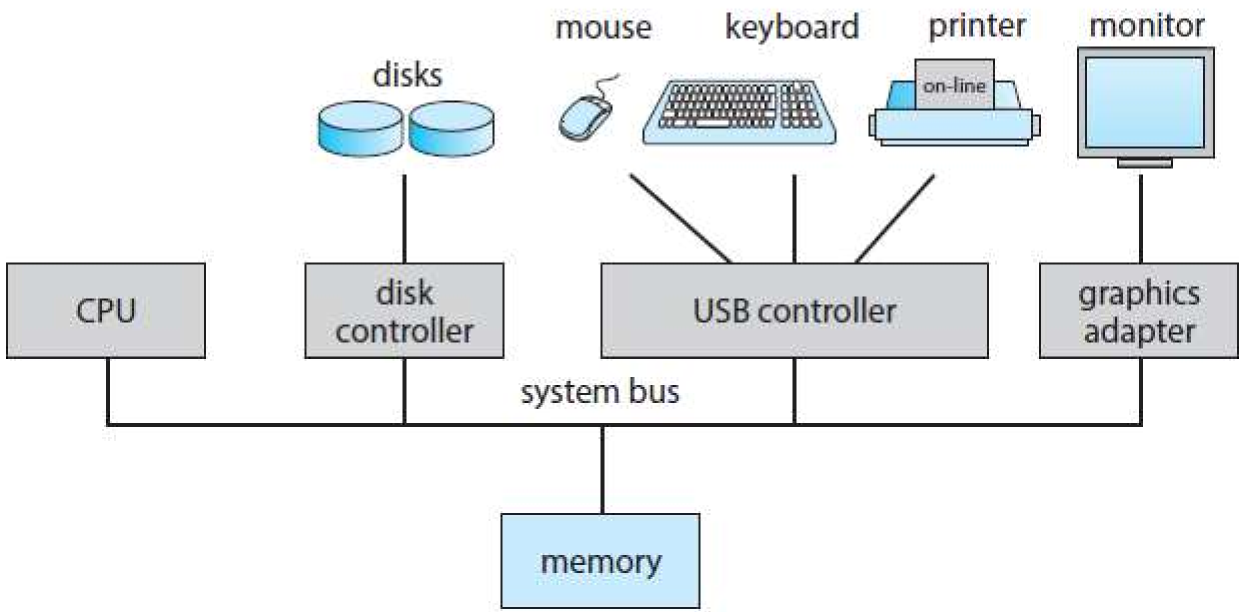


Figure 1.2 A typical PC computer system.

Computer-System Organization

- I/O devices and the CPU can execute concurrently
- Each device controller is in charge of a particular device type
- Each device controller has a local buffer
- CPU with a device driver moves data from/to main memory to/from local buffers
- Device controller informs CPU that it has finished its operation (such as “write completed successfully”) by causing an interrupt

Interrupt

- OS is **interrupt** driven by hardware
  - Interrupt: a signal sent to the processor to notify a (critical) event
    - E.g., a mouse button has been clicked
  - Hardware and software may trigger an interrupt

- Software-generated interrupts are **trap** or **exception**
  - Trap: software request for OS service (**system call** like file open)
  - Exception: software error (division by zero or invalid memory access

Interrupt

- On interrupts
  - Finish executing the current instruction
  - Save the state of the user program
  - Transfer execution to a fixed location (which contains the starting address where the service routine for the interrupt is located)
  - Execute the interrupt service routine
  - Restore the state of the user program
  - Resume to execute the user program where it was interrupted

Interrupt

- Interrupts must be handled quickly, as they occur very frequently.
- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector** (a table of pointers to interrupt routines), which contains the addresses of all the service routines
  - The interrupt routine is called indirectly through the table by a unique number (index).

vector number	description
0	divide error
1	debug exception
2	null interrupt
3	breakpoint
4	INTO-detected overflow
5	bound range exception
6	invalid opcode
7	device not available
8	double fault
9	coprocessor segment overrun (reserved)
10	invalid task state segment
11	segment not present
12	stack fault
13	general protection
14	page fault
15	(Intel reserved, do not use)
16	floating-point error
17	alignment check
18	machine check
19–31	(Intel reserved, do not use)
32–255	maskable interrupts

Figure 1.5 Intel processor event-vector table.